CS 5316 – NLP Theory and Applications

Project Ideas & Guidelines

Fall 2026

1. Project Overview

The strongest projects should go beyond simply training a classifier on a dataset. Each project should formulate a research question, establish meaningful baselines, perform a controlled intervention (for example, prompting, fine-tuning, retrieval, data augmentation, or translation), and analyze why the method succeeds or fails.

- Projects may use Urdu, Punjabi, Sindhi, Pashto, Saraiki, Balochi, or other low-resource languages, together with English or other comparison languages.
- Projects may focus on low-resource NLP, foundation-model behavior, LLM adaptation, multilinguality, RAG, bias, hallucination, or emerging applications.
- Where feasible, compare more than one model or method and report both quantitative and qualitative results.
- The final project should produce an evidence-based conclusion, not only a software demonstration.

2. Project Tracks

Track	Focus
Track A — Low-Resource NLP	Urdu/Punjabi/Sindhi/Pashto and other low-resource settings; data scarcity; transfer learning; PEFT; instruction tuning; synthetic data.
Track B — Foundation-Model Behavior	Multilinguality, bias, hallucination, cultural knowledge, language-dependent reasoning, and robustness.
Track C — LLM Adaptation	Prompting, RAG, LoRA/QLoRA, instruction tuning, data selection, synthetic data, translation, and catastrophic forgetting.

3. Project Options

The ideas below are organized by theme. A selected idea is a starting point rather than a complete specification. Each group should define its dataset, models, baselines, intervention, evaluation metrics, and experimental controls.

3.1 Low-Resource Language NLP

These projects are particularly appropriate for Urdu, Punjabi, Sindhi, Pashto, Saraiki, Balochi, or other under-resourced languages.

1. Urdu LLM Adaptation

Research question	Suggested design / tasks
How much can a general LLM be improved for Urdu with a small amount of domain-specific data?	Compare a general model with an adapted version using a controlled amount of domain-specific data. Possible tasks: classification, QA, summarization

2. Urdu vs. English Prompting

Research question	Suggested design / tasks
Does an LLM perform differently when the same task is expressed in Urdu versus English?	Keep the task and underlying content comparable while varying the prompt language. Possible tasks: QA, reasoning, sentiment

3. Cross-Lingual Transfer to Urdu

Research question	Suggested design / tasks
How effectively does knowledge learned from English transfer to Urdu?	Train or prompt using English and evaluate how much performance transfers to Urdu. Possible tasks: NLI, sentiment, topic classification

4. Urdu Data Augmentation

Research question	Suggested design / tasks
Which augmentation strategies produce the largest gains with limited training data?	Compare alternative augmentation strategies under the same training-data budget. Possible tasks: back-translation, paraphrasing, LLM-generated data

5. Urdu Instruction Tuning

Research question	Suggested design / tasks
Can a small instruction-tuning dataset substantially improve an LLM's Urdu instruction-following?	Construct or obtain a small instruction dataset and compare the base and instruction-tuned models. Possible tasks: QA, summarization, classification

6. Few-Shot Urdu NLP

Research question	Suggested design / tasks
How many examples are actually required before an LLM becomes useful for a low-resource task?	Compare performance as the number of demonstrations increases. Possible tasks: 0/1/5/10/50-shot experiments

7. Code-Switched Urdu-English NLP

Research question	Suggested design / tasks
How badly do NLP models degrade on naturally occurring Urdu-English code-switching?	Compare monolingual and code-switched inputs using naturally occurring or carefully constructed examples. Possible tasks: sentiment, toxicity, intent

8. Roman Urdu NLP

Research question	Suggested design / tasks
Can models handle Urdu written in Roman script as effectively as Urdu script?	Compare Roman Urdu and Urdu-script versions of the same or comparable tasks. Possible tasks: sentiment, intent, classification

9. Urdu Dialect Identification

Research question	Suggested design / tasks
Can models distinguish regional or dialectal variation in Urdu?	Build or use a dialect-labeled dataset and compare performance across dialects. Possible tasks: dialect classification

10. Punjabi NLP

Research question	Suggested design / tasks
How well do multilingual LLMs handle Punjabi compared with Urdu and English?	Use matched tasks and evaluation protocols across languages. Possible tasks: QA, sentiment, classification

11. Urdu–Punjabi Transfer

Research question	Suggested design / tasks
Does linguistic similarity between Urdu and Punjabi facilitate transfer learning?	Compare transfer settings and identify where linguistic similarity helps or fails to help. Possible tasks: classification, NER

12. Extremely Low-Resource NER

Research question	Suggested design / tasks
How much annotated data is necessary to build useful NER for a language with little labeled data?	Construct progressively larger labeled training sets and measure the data-efficiency curve. Possible tasks: PER/ORG/LOC NER

Featured project — "How much Urdu data is enough?"

Create progressively larger training sets, for example 100, 500, 1K, 5K, and 10K examples, and compare:

- zero-shot LLM
- few-shot prompting
- linear/TF-IDF baseline
- multilingual transformer
- parameter-efficient fine-tuning

The final result should be a data-efficiency curve for Urdu NLP, together with an analysis of where additional data stops providing substantial benefit.

3.2 Post-Training and Instruction Tuning for Low-Resource Languages

13. LoRA vs. Full Fine-Tuning for Urdu

Research question	Suggested design / tasks
How much does parameter-efficient fine-tuning help a low-resource language?	Compare zero-shot use, prompting, LoRA/QLoRA, and full fine-tuning; analyze quality, training cost, and catastrophic forgetting. Possible tasks: accuracy/F1, training cost, catastrophic forgetting

14. Instruction Tuning with Synthetic Urdu Data

Research question	Suggested design / tasks
Can synthetic data substitute for expensive human annotation?	Compare the original model, human-data fine-tuning, synthetic-data fine-tuning, and a mixture of human + synthetic examples. Possible tasks: instruction following, task performance

15. Does English Instruction Tuning Improve Urdu?

Research question	Suggested design / tasks
Does English instruction tuning transfer to Urdu, and how does	Start with a multilingual foundation model; fine-tune on English

it compare with Urdu instruction tuning?	instruction data; evaluate Urdu; then compare with Urdu instruction tuning. Possible tasks: cross-lingual transfer, language interference

16. Instruction Tuning and Catastrophic Forgetting

Research question	Suggested design / tasks
Does improving Urdu damage performance in English or other languages?	Measure performance before and after Urdu fine-tuning on Urdu, English, and another multilingual language. Possible tasks: multilingual evaluation

17. Which Urdu Examples Are Most Valuable?

Research question	Suggested design / tasks
Can 1,000 carefully selected Urdu examples outperform 10,000 random examples?	Compare random, diverse, high-quality, difficult, and LLM-selected examples under controlled data budgets. Possible tasks: data selection, low-resource fine-tuning

18. Instruction Tuning vs. Retrieval Augmentation

Research question	Suggested design / tasks
For a low-resource language, when is fine-tuning better than RAG, and when is a combination better?	Compare fine-tuning, RAG, and fine-tuning + RAG on domain-specific QA. Possible tasks: domain-specific QA

3.3 Foundation-Model Bias Studies

19. Gender Bias in LLMs

Research question	Suggested design / tasks
Do model predictions change systematically with gender-associated names, occupations, or roles?	Use controlled prompts such as “The engineer said that ___ solved the problem” and compare associations. Possible tasks: occupations, leadership positions, academic fields

20. Cultural Bias in Foundation Models

Research question	Suggested design / tasks
Does a model give culturally different answers depending on the language or cultural framing?	Compare culturally equivalent scenarios expressed in English, Urdu, Arabic, and Hindi. Possible tasks: controlled cultural comparisons

21. Geographic Bias

Research question	Suggested design / tasks
Are some countries represented substantially better than others?	Ask comparable questions involving Pakistan, India, the US, the UK, the Middle East, and Africa; measure factuality, sentiment, stereotypes, and information richness. Possible tasks: factuality, sentiment, stereotypes

22. Socioeconomic Bias

Research question	Suggested design / tasks
Do socioeconomic descriptions change model judgments about people?	Use matched prompts describing wealthy versus poor backgrounds and compare judgments. Possible tasks: intelligence, honesty, competence, education, employment, criminality

23. Name-Based Bias

Research question	Suggested design / tasks
Do names associated with different countries, genders, or ethnic/cultural groups produce systematically different model responses?	Use matched names and identical prompts, then quantify output differences. Possible tasks: controlled matched-name evaluation

24. Bias Amplification Through Fine-Tuning

Research question	Suggested design / tasks
Can task-specific fine-tuning unintentionally amplify social biases?	Measure bias before and after fine-tuning on a task dataset such as news, and test whether selected associations become stronger. Possible tasks: bias measurement, task fine-tuning

3.4 Bias Mitigation Projects

25. Can Prompting Mitigate Bias?

Research question	Suggested design / tasks
Can prompt design reduce measured bias without substantially harming useful capabilities?	Compare a normal prompt, fairness-aware prompting, explicit anti-stereotype instructions, and—where appropriate—reasoning-oriented prompting. Possible tasks: bias measurement, task performance

26. Bias Mitigation Through Fine-Tuning

Research question	Suggested design / tasks
Can a small debiasing dataset reduce bias without sacrificing useful model capabilities?	Compare ordinary task fine-tuning with debiasing fine-tuning and examine task performance, bias, and unintended side effects. Possible tasks: task performance, bias, side effects

27. Bias Mitigation Through Counterfactual Data

Research question	Suggested design / tasks
Does counterfactual augmentation reduce gender bias?	Create paired examples such as “The woman was a nurse” / “The man was a nurse” and evaluate the effect of the paired data. Possible tasks: counterfactual augmentation

28. Bias Across Languages After Mitigation

Research question	Suggested design / tasks
Does debiasing in one language transfer to another language?	Debias in English and test Urdu/Hindi/Arabic, or debias in Urdu and test English. Possible tasks: cross-lingual bias evaluation

3.5 Hallucination and Factuality

29. Hallucination in Low-Resource Languages

Research question	Suggested design / tasks
Are LLMs more prone to hallucination when answering in low-resource languages?	Ask the same factual questions in English, Urdu, Punjabi, and Hindi and compare hallucination rate, citation quality,

	confidence, and answer completeness. Possible tasks: factual QA

30. Does Translation Reduce Hallucination?

Research question	Suggested design / tasks
Does translating a low-resource-language query into English improve factuality?	Compare an Urdu→English→answer→Urdu pipeline with direct Urdu model responses. Possible tasks: factual QA, translation

31. LLM Knowledge of Pakistani Entities

Research question	Suggested design / tasks
How well do LLMs know Pakistani people, universities, cities, historical events, organizations, literature, and local terminology?	Create a benchmark of Pakistani entities and compare performance across languages and/or models. Possible tasks: knowledge evaluation, factual QA

3.6 Novel NLP Applications

32. Urdu Academic-Paper Assistant

Research question	Suggested design / tasks
Can an Urdu-oriented academic assistant provide useful summaries, explanations, questions, bilingual summaries, and retrieval?	Build a system that summarizes papers, explains terminology in Urdu, generates questions, produces bilingual summaries, and retrieves related papers. Compare ordinary RAG with multilingual RAG. Possible tasks: summarization, explanation, question generation, retrieval

33. Urdu Legal-Document Assistant

Research question	Suggested design / tasks
Does RAG reduce hallucination compared with direct LLM prompting for legal-document QA?	Build a retrieve→summarize→answer pipeline over legal documents and compare it with direct prompting. Possible tasks: document retrieval, summarization, QA

34. Urdu Educational Tutor

Research question	Suggested design / tasks
Which adaptation strategy produces the most useful Urdu educational tutor?	Compare a generic LLM, system-prompt tutor, RAG tutor, and fine-tuned tutor; evaluate factuality and pedagogical quality. Possible tasks: education, RAG, fine-tuning

35. Personalized Feedback on Student Writing

Research question	Suggested design / tasks
Are LLM-generated writing evaluations consistent across languages?	For Urdu/English essays, generate grammar, organization, vocabulary, and argument-quality feedback; compare quality and consistency. Possible tasks: writing feedback

36. Automatic Generation of Urdu Quizzes

Research question	Suggested design / tasks
Can LLMs reliably generate useful Urdu quizzes from lecture material?	Generate MCQs, short-answer questions, and conceptual questions; evaluate factual correctness, difficulty, ambiguity, and answerability.

	Possible tasks: question generation, educational NLP

37. Fake-News Detection Across Languages

Research question	Suggested design / tasks
How do classical, multilingual-transformer, prompting, and fine-tuned-LLM approaches compare for multilingual fake-news detection?	Compare TF-IDF + SVM, a multilingual transformer, LLM prompting, and a fine-tuned LLM; include cross-lingual transfer such as train English/test Urdu. Possible tasks: fake-news detection, cross-lingual transfer

38. Misinformation Translation

Research question	Suggested design / tasks
Does machine translation preserve the linguistic cues needed for misinformation detection?	Compare English misinformation→Urdu translation→classifier with Urdu misinformation→English translation→classifier. Possible tasks: misinformation detection, machine translation

3.7 Multilingual and Cross-Lingual Projects

39. Language Switching in Multilingual LLMs

Research question	Suggested design / tasks
Does model behavior change when the same semantic content is expressed in different languages?	Compare language-conditioned performance on accuracy, verbosity, reasoning, hallucination, and refusal behavior. Possible tasks: multilingual evaluation

40. Does Language Affect LLM Reasoning?

Research question	Suggested design / tasks
Is reasoning capability invariant across languages?	Use mathematically or logically equivalent problems in English, Urdu, Hindi, and Arabic and compare performance. Possible tasks: reasoning evaluation

41. Translation as a Universal Adapter

Research question	Suggested design / tasks
How does translation-based task transfer compare with multilingual task models and Urdu fine-tuning?	Compare Urdu→English→task model, Urdu→multilingual task model, and an Urdu fine-tuned model. Possible tasks: cross-lingual transfer

42. Cross-Lingual Prompt Transfer

Research question	Suggested design / tasks
Does prompt engineering transfer across languages?	Create prompts optimized for English and compare literal translation, culturally adapted translation, and manually optimized Urdu prompts. Possible tasks: prompt engineering, cross-lingual transfer

3.8 Retrieval-Augmented Generation

43. Does RAG Help Low-Resource Languages?

Research question	Suggested design / tasks
Does RAG improve factuality for low-resource-language questions?	Build an Urdu knowledge base and compare LLM alone, English RAG, Urdu RAG, and bilingual RAG. Measure factuality and retrieval quality separately.

	Possible tasks: RAG, factual QA

44. RAG Retrieval Bias

Research question	Suggested design / tasks
Can retrieval introduce bias even when the underlying LLM is comparatively unbiased?	Investigate whether retrieval disproportionately returns information about certain countries, groups, or perspectives. Possible tasks: retrieval evaluation, bias

45. Multilingual RAG

Research question	Suggested design / tasks
Should an Urdu question retrieve Urdu documents only?	Compare Urdu-only retrieval, English-only retrieval, and multilingual retrieval. Possible tasks: multilingual retrieval, QA

4. Recommended Project Tiers

These recommendations are intended for groups of 2–3 students working over approximately 2–4 weeks; they are not restrictions.

Tier A — Strong Research Projects

1. How much Urdu data is enough for fine-tuning?
2. LoRA vs. prompting for Urdu NLP
3. Synthetic Urdu instruction data vs. human data
4. Hallucination rates across English and Urdu
5. Gender/cultural bias across languages
6. Bias amplification after fine-tuning
7. Can counterfactual augmentation reduce bias?
8. English-to-Urdu cross-lingual transfer
9. Does RAG improve Urdu factuality?
10. Roman Urdu vs. Urdu-script NLP

Tier B — More Ambitious Projects

11. Multilingual RAG for Pakistani information
12. Urdu instruction tuning
13. Cross-lingual bias mitigation
14. Language-dependent reasoning in LLMs
15. Data selection for low-resource fine-tuning
16. Synthetic data quality for low-resource languages
17. Cross-lingual prompt transfer
18. Fine-tuning vs. RAG vs. fine-tuning + RAG

5. Required Common Project Structure

Part 1 — Research Question: State a precise, testable hypothesis. Example: “Parameter-efficient fine-tuning provides most of the benefit of full fine-tuning for Urdu sentiment classification while requiring substantially less computation.”

Part 2 — Baselines: Use at least 2–3 meaningful baselines where feasible, such as TF-IDF + SVM, a multilingual Transformer, and a zero-shot LLM.

Part 3 — Intervention: Apply at least one controlled intervention: LoRA/QLoRA, instruction tuning, prompting, RAG, synthetic data, data augmentation, translation, or bias mitigation.

Part 4 — Evaluation: Report more than one number. Depending on the project, include Accuracy/F1, performance by class, performance by language, robustness, error analysis, and computational cost.

Part 5 — Ablation: Vary an important factor systematically, for example 100/500/1K/5K training examples or English-only/Urdu-only/bilingual data.

Part 6 — Error and Qualitative Analysis: Inspect a meaningful sample of errors (the source recommends roughly 20–50) and categorize the failure modes.

Part 7 — Research Conclusion: Answer: "What did we learn that we did not know before running the experiment?" The conclusion should be evidence-based and tied to the experiments.

6. Project Planning Checklist

- Research question and hypothesis are precise and measurable.
- Dataset(s) and train/dev/test protocol are clearly specified.
- At least two meaningful baselines are identified, where feasible.
- The intervention is controlled so that comparisons are interpretable.
- Evaluation metrics match the research question.
- Ablations test which component is responsible for observed gains or losses.
- Results include quantitative analysis and representative qualitative examples.
- Limitations, data quality issues, and possible sources of bias are discussed.
- Computational resources and practical costs are reported when relevant.
- The final conclusion answers the research question rather than merely describing the implementation.

7. Recommended Overarching Course Theme

A particularly coherent class-wide theme is "Adapting and Evaluating Foundation Models for Low-Resource Languages." Different groups can investigate Urdu adaptation, synthetic data, bias, hallucination, RAG, cross-lingual transfer, instruction tuning, or fairness.

A shared evaluation framework can allow the class to contribute results to a common low-resource-language benchmark while still giving each group a distinct research question.

8. Using This Handout

Select a project idea as a starting point, then refine it into a specific research question and experimental plan. Projects should be scoped to the available time, data, compute, and group size. Any substantial change in scope should be discussed with the instructor before implementation.

Prepared by reorganizing and standardizing the supplied NLP projects document while retaining its substantive project ideas and research directions.